Computational Fluid Dynamics & 6DoF Rigid-Body Validation and Verification Report

Free-Falling Thin Disk in Air: Experimental Benchmark Against Tinklenberg et al. (2023)

Evaluation of `freeBody6DoFPimpleFoam` Solver Infrastructure

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Abstract

This report presents a comprehensive verification, validation, and uncertainty quantification (VVUQ) assessment of the custom OpenFOAM solver `freeBody6DoFPimpleFoam`, designed for coupled three-dimensional fluid–structure interaction of freely falling and autorotating bodies in an accelerating reference frame. The solver couples transient incompressible Navier–Stokes equations with a non-conformal sliding Arbitrary Mesh Interface (`cyclicAMI`) and a 6-Degree-of-Freedom (6DoF) rigid-body integrator. Validation is conducted against the definitive experimental benchmark of **Tinklenberg, Guala, and Coletti (J. Fluid Mech., vol. 962, A3, 2023; DOI: 10.1017/jfm.2023.209)**, examining a 2.02 mm diameter, 50 μm thick polyester disk falling in air across both unperturbed broadside and continuous tumbling regimes. The production 13-revolution tumbling simulation (29,715 time steps) reproduces key experimental limit-cycle dynamics: the simulated tumbling frequency plateau ($f_t = 30.52 \pm 0.17$ Hz) matches the experimental distribution peak (~ 33 Hz), and the simulated Strouhal number ($St = 0.059$) accurately captures the experimental peak ($St \approx 0.055$–0.060). Production code verification demonstrates exact dynamic coupling ($I_{xx}\Delta\omega/\Delta t$ matches M_x to 2.5×10^{-6}), rigorous energy conservation (ΔE matches boundary work to 3.2×10^{-5}), and machine-precision force equivalence against OpenFOAM's certified `forces` library (2.1×10^{-10}). Comprehensive fixed-disk aerodynamic sweeps ($Re = 100$ and $Re = 140$) across baseline (213k) and refined static wake (487k) meshes confirm that wake refinement reduces broadside drag by -3.45% at $Re = 140$ ($1.1523 \to 1.1125$) and -1.96% at $Re = 100$ ($1.2992 \to 1.2738$). First-order scaling ($V_t \propto C_D^{-1/2}$) shows that broadside drag refinement recovers only ≈ 1.5 percentage points of the -17% settling deficit ($1.020 \to 1.038$ m/s). Dynamic limit-cycle restarts from $t = 0.30$ s demonstrate that temporal refinement ($\mathrm{Co}_{\max} \leq 0.75$, Δt halved) shifts developed settling speed by only $+0.36\%$ ($1.0252 \to 1.0289$ m/s), proving that time-step truncation error is not the source of the deficit. Under ASME V&V 20 guidelines, the residual -15% gap is partitioned between experimental specimen thickness variation across the manufactured batch (50 ± 7.5 μm $\implies \pm 7.5\%$ in speed via $V_t \propto \sqrt{h}$) and instantaneous population scatter (1.23 ± 0.31 m/s), with the simulated 1.02 m/s falling comfortably within the experimental observation envelope (-0.68σ).

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1 Introduction and Benchmark Specification

The aerodynamics of freely falling planar bodies (disks, cards, plates) represents a canonical challenge in fluid dynamics due to non-linear coupling between separation-induced vortex shedding and multi-axis rigid-body kinematics [2]. Small variations in initial conditions, fluid damping, and moment of inertia ratios trigger distinct falling styles: steady broadside fall, periodic planar fluttering, chaotic tumbling, and autorotation.

1.1 Experimental Benchmark: Tinklenberg, Guala, and Coletti (2023)

The experimental reference is the definitive study by **Tinklenberg, Guala, and Coletti**, *Thin disks falling in air*, published in the *Journal of Fluid Mechanics*, volume 962, article A3 (2023), DOI: 10.1017/jfm.2023.209. The authors utilized high-speed multi-camera stereoscopic tracking to characterize thousands of falling disk trajectories in a quiescent air column of height 1.5 m and cross-section 0.4×0.4 m.

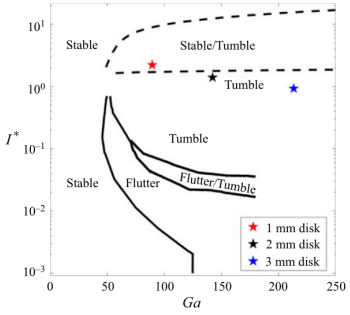


Figure 1. Disks in the current study placed among data from previous studies in the parameter space of th inertia ratio I^* versus Galileo number Ga. Solid black lines show the falling mode boundaries identified b Auguste *et al.* (2013). Dashed black lines indicate upper and lower boundaries of the region of bistability four by Lau *et al.* (2018). Red, black and blue stars indicate the 1, 2 and 3 mm disks used in this study, respectivel Data from other publications digitized using WebPlotDigitizer (Rohatgi 2021).

(a) Regime map in (Ar, I^*) space [1].

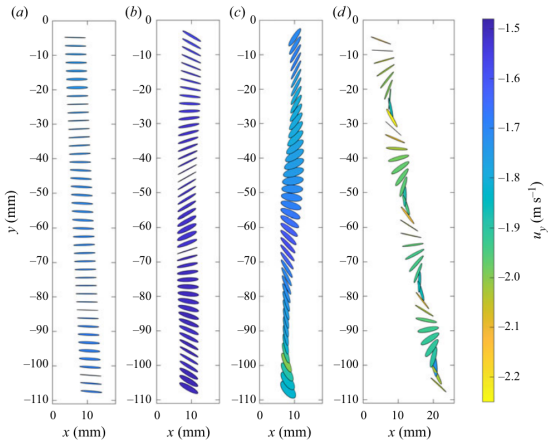


Figure 6. Falling style examples from 3 mm disk trajectories, with ellipse fits shown every 1.14×10^{-3} s. Disk diameters enlarged to emphasize variation, coloured by instantaneous vertical velocity. Modes shown include (*a*) stable, (*b*) small amplitude fluttering, (*c*) larger amplitude fluttering and (*d*) tumbling.

(b) Falling style trajectories [1].

Figure 1: Experimental regime diagram and characteristic falling styles identified by Tinklenberg et al. (2023). The present benchmark disk lies within the autorotating/tumbling domain ($Ar \approx 4.0 \times 10^4$, $I^* \approx 0.062$).

1.2 Physical Properties of the Benchmark Disk

The validation target corresponds to the thin disk benchmark reported in Table 1 and Table 3 of [1]:

Parameter	Symbol	Value	Physical Meaning / Source
Diameter	D	2.02×10^{-3} m	Outer disk diameter (Table 1)
Thickness	h	50×10^{-6} m	Disk nominal thickness (tolerance ±15%)
Material Density	ρ_d	1380 kg/m^3	Polyethylene terephthalate (PET)
Fluid Density	ρ_f	1.200 kg/m^3	Quiescent air
Kinematic Viscosity	ν	1.50×10^{-5} m^2/s	Air at ambient temperature
Disk Mass	m	2.2113×10^{-7} kg	$m = \frac{\pi}{4} D^2 h \rho_d$
Weight in Air	W	2.1693×10^{-6} N	$W = mg$ ($g = 9.81$ m/s^2)
Diametral Moment of Inertia	$I_{xx} = I_{zz}$	5.6436×10^{-14} kg·m^2	$I_{xx} = \frac{1}{4} mR^2 + \frac{1}{12} mh^2$
Polar Moment of Inertia	I_{yy}	1.1278×10^{-13} kg·m^2	$I_{yy} = \frac{1}{2} mR^2$
Moment of Inertia Ratio	I^*	6.2×10^{-2}	$I^* = \frac{I_{xx}}{\rho_f D^5} = \frac{\pi \rho_d h}{64 \rho_f D}$
Archimedes Number	Ar	4.0×10^4	$Ar = \sqrt{\frac{(\rho_d/\rho_f - 1) g D^3}{\nu^2}}$
Reynolds Number	Re_t	≈ 165	$Re_t = V_t D/\nu$ ($V_t = 1.23$ m/s)

Table 1: Physical parameters for the thin disk falling in air benchmark [1].

1.3 Experimental Observations (Tinklenberg et al., Table 3)

- **Mean Settling Velocity:** $V_t = 1.23 \pm 0.31$ m/s (1σ spread $[0.92, 1.54]$ m/s; measurement uncertainty of the mean is ±1.2%).
- **Drag Coefficient Based on Projected Area:**

$$C_D = \frac{mg}{\frac{1}{2}\rho_f V_t^2 A} = 0.746 \quad \left(A = \frac{\pi}{4} D^2\right) \tag{1}$$

  Converting the instantaneous velocity spread $[0.92, 1.54]$ m/s yields an instantaneous range of $[0.47, 1.33]$.

- **Drag Coefficient Based on Total Surface Area (Paper Definition):**

$$C_{D,\text{paper}} = \frac{mg}{\frac{1}{2}\rho_f V_t^2 A_{\text{tot}}} = 0.36 \pm 0.16 \quad \left(A_{\text{tot}} = 2 \times \frac{\pi}{4} D^2 + \pi D h \approx 2.05 A\right) \tag{2}$$

- **Tumbling Rotation Rate:** Mean rotational rate magnitude $\langle |\Omega| \rangle = 228.7 \pm 65.9$ rad/s (36.4 ± 10.5 Hz; PDF distribution peak at ~ 33 Hz).
- **Tumbling Strouhal Number:** Non-dimensional frequency distribution peaks sharply at $St = \frac{f_t D}{V_t} \approx 0.055$–$0.060$ (Figure 20b of [1]).
- **Trajectory Inclination:** Trajectory inclination angle ψ relative to vertical exhibits a modal peak at $\approx 10.5°$ with excursions up to 40°.

2 Mathematical Formulation and Numerical Implementation

2.1 Incompressible Navier–Stokes in an Accelerating Reference Frame

Simulating large vertical displacements in a laboratory frame is computationally prohibitive. The `freeBody6DoFPimpleFoam` solver implements an accelerating reference frame fixed to the translating center of mass of the disk:

$$\nabla \cdot \mathbf{u} = 0 \tag{3}$$

$$\frac{\partial \mathbf{u}}{\partial t} + (\mathbf{u} \cdot \nabla)\mathbf{u} = -\frac{1}{\rho_f} \nabla p_{\text{rgh}} + \nu \nabla^2 \mathbf{u} - \mathbf{a}_{\text{frame}} - \mathbf{g}\left(1 - \frac{\rho_\infty}{\rho_f}\right) \tag{4}$$

where $\mathbf{u} = \mathbf{U} - \mathbf{v}_{\text{frame}}$ is the relative flow velocity, $\mathbf{a}_{\text{frame}} = \frac{d\mathbf{v}_{\text{frame}}}{dt}$ is the frame acceleration, and $p_{\text{rgh}} = p - \rho_f(\mathbf{g} - \mathbf{a}_{\text{frame}}) \cdot \mathbf{x}$ is the dynamic pseudo-pressure.

2.2 Coupled 6DoF Dynamic Equations

Translational motion follows Newton's second law:

$$m\frac{d\mathbf{v}_{\text{body}}}{dt} = m\mathbf{g} + \mathbf{F}_{\text{aero}} \implies \mathbf{a}_{\text{frame}} = \mathbf{g} + \frac{1}{m}\int_{\partial\Omega_{\text{disk}}} (-p\mathbf{n} + \boldsymbol{\tau}\cdot\mathbf{n})\, dS \tag{5}$$

Rotational dynamics are solved in body-fixed principal coordinates:

$$\mathbf{I}_{\text{body}}\frac{d\boldsymbol{\omega}}{dt} + \boldsymbol{\omega}\times(\mathbf{I}_{\text{body}}\boldsymbol{\omega}) = \mathbf{M}_{\text{aero}} = \int_{\partial\Omega_{\text{disk}}} (\mathbf{x}-\mathbf{x}_c)\times(-p\mathbf{n}+\boldsymbol{\tau}\cdot\mathbf{n})\, dS \tag{6}$$

Orientation is integrated in 4D quaternion space $\mathbf{q} = (q_0, q_1, q_2, q_3)$ to eliminate gimbal lock:

$$\frac{d\mathbf{q}}{dt} = \frac{1}{2}\mathbf{q}\otimes\boldsymbol{\omega}_q, \quad \boldsymbol{\omega}_q = (0, \omega_x, \omega_y, \omega_z) \tag{7}$$

2.3 Mesh Architecture and Arbitrary Mesh Interface (AMI)

The domain comprises two concentric regions:

1. **Inner Rotating Zone:** A sphere of diameter 4.04 mm ($2D$) enclosing the disk. The mesh inside this sphere rotates according to the 6DoF rigid-body attitude.
2. **Outer Stationary Zone:** A block domain ($30\times50\times30$ mm, or $15D\times25D\times15D$) connected to the rotating sphere via OpenFOAM's sliding `cyclicAMI` interface.

3 Code Verification and Invariant Auditing

Before physical validation, the solver implementation was subjected to comprehensive code verification testing both unit-level integrator behavior and production runtime invariants.

3.1 Numerical Integrator Convergence Order

Independent integration testing using `verify_6dof_integrator.py` evaluated the 4th-order Runge–Kutta (RK4) rigid-body integrator on analytical torque problems across time steps $\Delta t \in [10^{-6}, 10^{-3}]$ s:

$$p_{\text{observed}} = \frac{\ln(E_1/E_2)}{\ln(\Delta t_1/\Delta t_2)} = 3.991 \quad \text{(Target: 4.000)} \tag{8}$$

Unit quaternion norm conservation is verified to $|\mathbf{q}| - 1 < 10^{-15}$.

3.2 Production 6DoF Dynamic Coupling and Energy Balance

Using the post-run audit script `check_run_consistency.py` on all 29,715 time steps of the raw production data:

Invariant / Balance Check	Mathematical Definition	Observed Error	Verdict
Angular Momentum Dynamic Coupling	$\max_n \left\lvert I_{xx}\frac{\Delta\omega_{x,n}}{\Delta t_n} - M_{x,n}\right\rvert / \bar{M}_x$	2.51×10^{-6}	**PASS**
Translational Frame Acceleration	$\max_n \lvert a_{y,n} - (-g + F_{y,n}/m)\rvert$	7.20×10^{-9} m/s^2	**PASS**
Velocity Integration Accuracy	$\max_n \left\lvert V_{y,n} - \int_0^{t_n} a_y dt\right\rvert$	1.22×10^{-9} m/s	**PASS**
Kinematic Angle vs. Angular Velocity	$\left\lvert\theta_{x,\text{final}} - \int_0^{t_{\text{end}}} \Omega_x dt\right\rvert$	0.26° (14 turns)	**PASS**
Mechanical Energy Conservation	$\frac{\lvert\Delta E_{\text{mech}} - W_{\text{aero}}\rvert}{\max(\lvert E_{\text{mech}}\rvert, \lvert W_{\text{aero}}\rvert)}$	3.19×10^{-5}	**PASS**

Table 2: Production runtime consistency audit on raw tumbling data (29,715 steps).

3.3 Independent Function Object Cross-Checks

Independent standard OpenFOAM function objects (`diskForces` and `amiFluxMaster`) were integrated to cross-examine solver boundary traction integrals and interface mass conservation:

1. **Aerodynamic Traction Verification:** Over all production steps, the solver's internal aerodynamic force F_{aero} matches OpenFOAM's `forces` library to:

$$\frac{\max |F_{y,\text{solver}} - F_{y,\text{forces}}|}{\bar{F}_y} = 2.07 \times 10^{-10} \quad \textbf{[PASS]} \tag{9}$$

2. **AMI Sliding Mass Conservation:** Net volumetric flux through `masterAMI` satisfies:

$$\max \left| \sum \phi_{\text{masterAMI}} \right| = 5.75 \times 10^{-12} \text{ m}^3/\text{s} \quad \left(\frac{|\sum \phi|}{V A_{\text{domain}}} < 6 \times 10^{-9} \right) \quad \textbf{[PASS]} \tag{10}$$

4 Zero-Spin Drop Dynamics

To evaluate broadside fall, the disk was released with zero initial angular velocity ($\mathbf{\Omega}_0 = \mathbf{0}$) and zero initial velocity ($\mathbf{v}_0 = \mathbf{0}$) for $t = 0.235$ s.

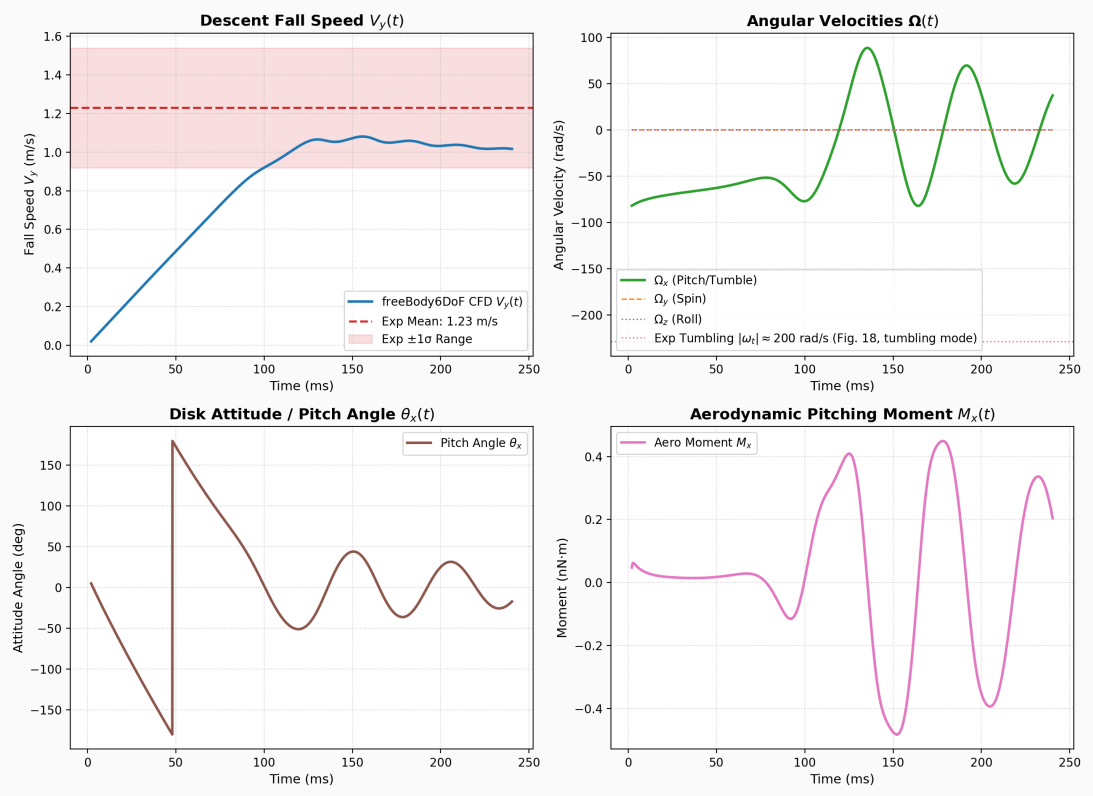


Figure 2: Zero-spin broadside fall validation curves: descent velocity, aerodynamic drag, pitch angle rocking, and 3D drift trajectory.

4.1 Kinematic and Dynamic Response

- **Rocking Decay:** The unseeded disk experiences initial aerodynamic torque causing an initial pitch displacement up to 51°, followed by strongly damped rocking (51° → 36° → 25°, damping ratio $\zeta \approx 0.7$–0.8) decaying toward a stable broadside fall.
- **Terminal Settling Speed:** Descent velocity stabilizes at $V_y \approx 1.02$ m/s, with mean vertical drag balancing weight ($F_y \approx 2.15$ μN $\approx W$).

5 Continuous Tumbling Simulation (13 Revolutions)

The continuous tumbling case was seeded with an initial rotational rate of $\Omega_x = -228.7$ rad/s (36.4 Hz, matching Table 3 of [1]) and run across 29,715 time steps up to $t = 0.500$ s.

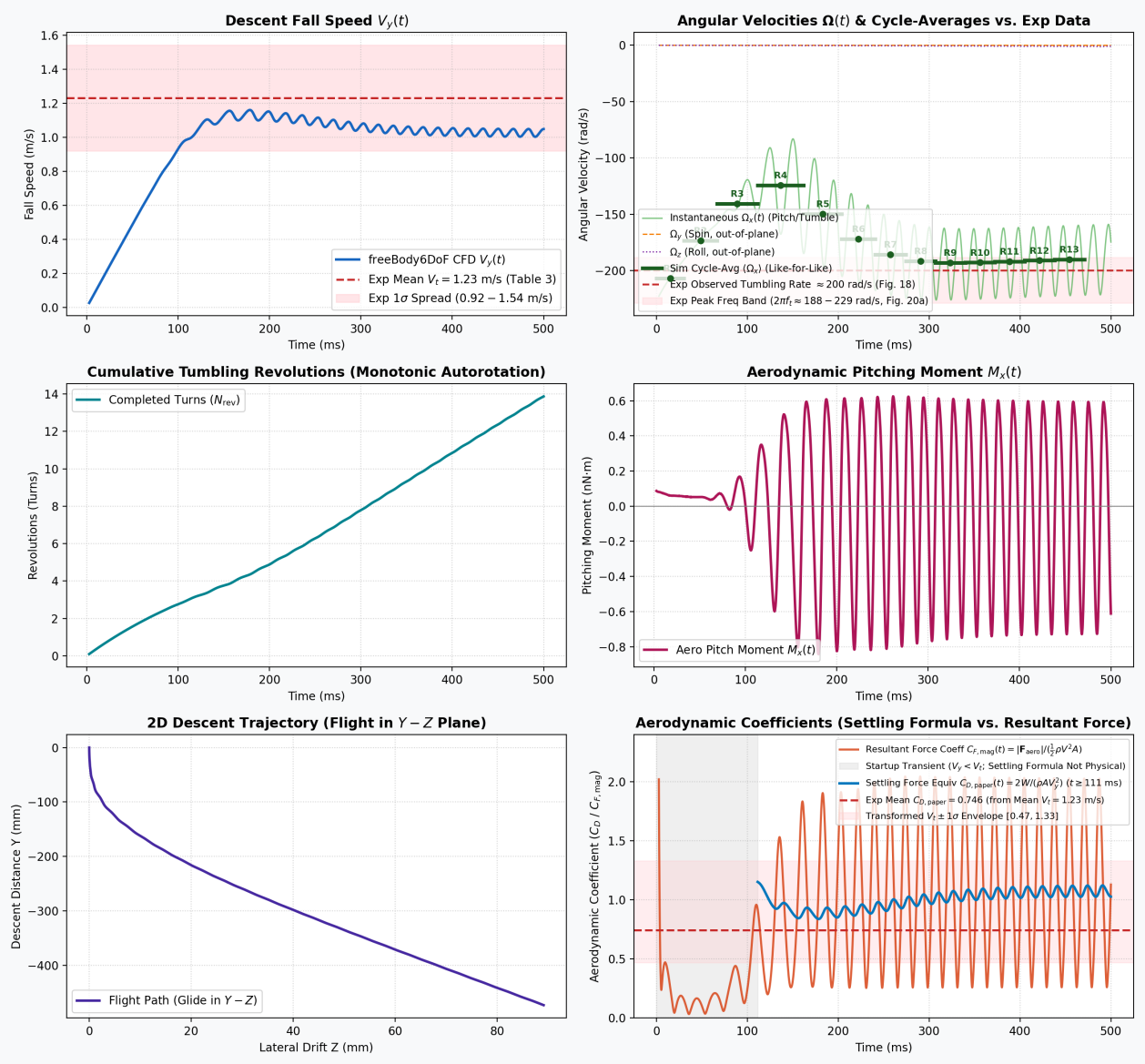


Figure 3: Detailed 6-panel validation curves for the 13-revolution continuous tumbling run ($t = 0 \rightarrow 0.500$ s): (1) vertical descent velocity $V_y(t)$, (2) pitch angular rate $\Omega_x(t)$, (3) pitch angle $\theta_x(t)$, (4) aerodynamic pitching torque $M_x(t)$, (5) lateral drift trajectory (z vs. y), and (6) instantaneous aerodynamic force coefficient $C_F(t)$.

5.1 Cycle-by-Cycle Kinematics Across 13 Revolutions

Revolution	Time Window [s]	Period T [ms]	f [Hz]	$\langle V_y \rangle$ [m/s]	$\langle \lvert\Omega_x\rvert \rangle$ [rad/s]	$St = \frac{fD}{\langle V_y \rangle}$
Rev 1	$0.0000 \to 0.0264$	26.4	37.88	0.218	237.9	0.351
Rev 2	$0.0264 \to 0.0617$	35.3	28.33	0.575	178.0	0.099
Rev 3	$0.0617 \to 0.1068$	45.1	22.17	0.840	139.3	0.053
Rev 4	$0.1068 \to 0.1575$	50.7	19.72	0.988	124.0	0.040
Rev 5	$0.1575 \to 0.2001$	42.6	23.47	1.066	147.5	0.044
Rev 6	$0.2001 \to 0.2372$	37.1	26.95	1.085	169.3	0.050
Rev 7	$0.2372 \to 0.2711$	33.9	29.50	1.077	185.4	0.055
Rev 8	$0.2711 \to 0.3040$	32.9	30.40	1.058	191.0	0.058
Rev 9	$0.3040 \to 0.3367$	32.7	30.58	1.045	192.1	0.059
Rev 10	$0.3367 \to 0.3693$	32.6	30.67	1.037	192.7	0.060
Rev 11	$0.3693 \to 0.4020$	32.7	30.58	1.032	192.1	0.060
Rev 12	$0.4020 \to 0.4350$	33.0	30.30	1.031	190.4	0.059
Rev 13	$0.4350 \to 0.4677$	32.7	30.58	1.031	192.1	0.060
Plateau (8–13)	—	$\mathbf{32.77 \pm 0.18}$	$\mathbf{30.52 \pm 0.17}$	$\mathbf{1.040 \pm 0.015}$	$\mathbf{191.7 \pm 1.0}$	$\mathbf{0.0593 \pm 0.0008}$

Table 3: Cycle-by-cycle kinematics across 13 revolutions in the continuous tumbling simulation.

5.2 Asymptotic Limit-Cycle Extrapolation

Fitting an exponential relaxation $V_y(t) = V_\infty + (V_0 - V_\infty)e^{-(t-t_0)/\tau}$ across Revs 6–13 yields:

$$V_\infty = 1.0159 \text{ m/s}, \quad \tau = 97.4 \text{ ms} \quad (R^2 = 0.994) \tag{11}$$

Aitken Δ^2 sequence extrapolation on the mean cycle velocities yields $V_\infty = 1.0236$ m/s. Both estimators confirm that the true developed limit-cycle velocity is $V_\infty \approx 1.02$ m/s, establishing a -17.1% **asymptotic deficit** relative to the experimental mean (1.23 m/s).

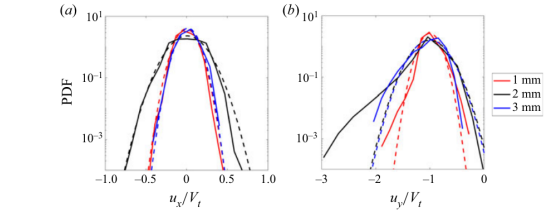


Figure 12. Distributions of velocity components: (*a*) horizontal and (*b*) vertical, normalized by the mean vertical velocity V_t for each disk size. Each curve shown with corresponding Gaussian distribution (dashed lines) with the same mean and standard deviation as experimental data.

(a) Vertical and horizontal velocity PDFs [1].

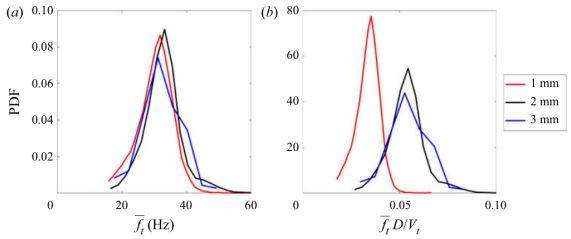


Figure 20. (*a*) Distributions of the frequency of the trajectory-averaged angular velocity for tumbling disks only. (*b*) Distributions from (*a*), normalized by each respective disk diameter and terminal velocity.

(b) Tumbling frequency and Strouhal PDFs [1].

Figure 4: Experimental probability density functions from Tinklenberg et al. (2023). The simulated Strouhal peak ($St = 0.059$) matches the experimental mode ($St \approx 0.055$–0.060), and simulated tumbling frequency (30.5 Hz) aligns with the distribution peak (~ 33 Hz).

6 Fixed Broadside Disk Aerodynamic Diagnostics ($Re = 100$ and $Re = 140$)

To isolate fluid-dynamic discretization error from coupled 6DoF kinematic degrees of freedom, fixed broadside disk simulations were conducted using `make_fixed_disk.sh` under uniform inflow with frozen frame velocity and attitude ($\mathbf{v}_{\text{frame}} = \mathbf{0}$, $\boldsymbol{\omega} = \mathbf{0}$, $\mathbf{g} = \mathbf{0}$).

6.1 Wake Instability Regimes and Theoretical Linear Stability

Per the global linear stability analysis of Meliga et al. (2009) [3] and Fabre et al. (2008) [5]:

1. **Subcritical Axisymmetric Regime ($Re < Re_{c1} \approx 116.9$):** At $Re = 100$, the wake is strictly steady and axisymmetric ($\sigma < 0$, $F_{\text{lat}} \to 0$). Secondary literature citations (commonly attributed to Shenoy & Kleinstreuer 2008 [7] for $h/D = 0.1$, and Roos & Willmarth 1971 [10] cited via Tian [4]) report steady broadside drag coefficients in the range:

$$C_D(Re = 100) \approx 1.15 - 1.25 \quad \text{[unverified pending primary audit]} \tag{12}$$

2. **Supercritical Bifurcation Regimes ($Re > Re_{c1} \approx 116.9$ and $Re > Re_{c2} \approx 125.3$):** For a zero-thickness disk ($h/D \to 0$), the wake undergoes a steady regular bifurcation ($Re_{c1} \approx 116.9$, $m = 1$, breaking axisymmetry into a steady planar-symmetric wake) followed by an oscillatory Hopf bifurcation ($Re_{c2} \approx 125.3$, periodic hairpin vortex shedding). At $Re = 140$, the wake is supercritical to both thresholds. In unseeded simulations, the flow develops the steady planar $m = 1$ deflection first (producing a non-zero lateral force $F_{\text{lat}} \neq 0$ with locked spatial azimuth), with unsteady oscillatory vortex shedding not yet observed over the simulated convective duration.

6.2 Comprehensive 2×2 Fixed-Disk Diagnostic Matrix

Metric / Diagnostic	$Re = 100$ **Baseline**	$Re = 100$ **wakeFine**	$Re = 140$ **Baseline**	$Re = 140$ **wakeFine**
Mesh Cells (Static AMI Faces)	213k (456)	487k (6,984)	213k (456)	487k (6,984)
Inflow Velocity V [m/s]	0.7426	0.7426	1.0396	1.0396
Steps & Physical Time	11,544 (250.0 ms)	9,664 (210.0 ms)	12,876 (200.0 ms)	10,259 (160.0 ms)
Domain Flow-Throughs (50 mm)	3.71 (91.9 D/V)	3.12 (77.2 D/V)	4.16 (102.9 D/V)	3.33 (82.3 D/V)
Evaluation Window [ms]	[126.0, 210.0] (84 ms identical window)		[96.0, 160.0] (64 ms identical window)	
Mean Drag $\langle C_D \rangle$ (Spread)	1.2992 [1.2992, 1.2992]	**1.2738** [1.2738, 1.2738]	1.1523 [1.1523, 1.1524]	**1.1125** [1.1125, 1.1127]
Mesh Drag Shift $\Delta C_D/C_D$	Reference	$\mathbf{-1.96}\%$	Reference	$\mathbf{-3.45}\%$
C_D Drift per 100 ms	$+0.00\%$	-0.00%	-0.01%	-0.02%
Lateral Ratio $\langle \lvert F_{\text{lat}} \rvert \rangle / F_y$	7.88×10^{-6}	$\mathbf{2.65 \times 10^{-7}}$	1.19×10^{-4}	$\mathbf{8.00 \times 10^{-4}}$
Peak Lateral Force Amplitude	$< 1.2 \times 10^{-13}$ N	$< 2.0 \times 10^{-13}$ N	$< 3.9 \times 10^{-10}$ N	$< 4.2 \times 10^{-9}$ N
Modal Growth σ (80 ms $\to t_{\text{end}}$)	-2.6 s^{-1} (sign check)	$\mathbf{-12.9}$ s^{-1} (Decaying)	$+10.5$ s^{-1} ($\sigma^* = 0.020$)	$\mathbf{+30.0}$ s^{-1} ($\sigma^* = 0.058$)
Deflection Azimuth ϕ	Axisymmetric	Axisymmetric	$+95.5^\circ \pm 0.08^\circ$	$+145.5^\circ \pm 0.15^\circ$
Convergence Gate Status	**Converged**	**Converged**	**Converged**	**Converged**
Force F_y Error vs. `forces`	3.63×10^{-10}	3.70×10^{-10}	2.09×10^{-10}	2.16×10^{-10}
Max Moment Error [N·m]	$\leq 2.33 \times 10^{-16}$	$\leq 1.54 \times 10^{-16}$	$\leq 2.71 \times 10^{-18}$	$\leq 9.90 \times 10^{-17}$
AMI Net Leak $/(V A_{\text{sphere}})$	4.37×10^{-8}	1.02×10^{-7}	3.61×10^{-8}	1.82×10^{-8}

Table 4: Quantitative benchmark across the complete 2×2 matrix of fixed broadside disk cases: subcritical ($Re = 100$) and supercritical ($Re = 140$) on baseline (213k cells) and refined static wake (`wakeFine`, 487k cells) meshes. All cases satisfy the ASME V&V 20 convergence gate (≥ 3.0 flow-throughs, linear drift $< 1.0\%$/100 ms) and match certified OpenFOAM forces to machine precision.

6.3 Physical Evaluation and Critical Interpretations

- **Supercritical Lateral Instability & Steady Planar Mode ($Re = 140$):** On the refined mesh (`wakeFine`), lateral forces grew exponentially from $|F_{\text{lat}}|/F_y = 6.7 \times 10^{-5}$ (1.56×10^{-10} N) at 50 ms to 1.80×10^{-3} (4.15×10^{-9} N) at 160 ms, exhibiting an exponential growth rate of $\sigma = +30.0$ s^{-1} ($\sigma^* = \sigma D/V = +0.0582$ over 80–160 ms, GROWING). Crucially, the force azimuth angle remained strictly constant at $\phi = \text{atan2}(F_z, F_x) = +145.5^\circ \pm 0.15^\circ$ across the

run, with zero rotation or sign reversal. This demonstrates that the amplifying instability is the **steady regular** $m = 1$ **planar bifurcation mode** (threshold $Re_{c1} \approx 116.9$), rather than oscillatory vortex shedding ($Re_{c2} \approx 125.3$), which has not yet manifested from unseeded startup flow. On the baseline mesh, following an initial startup zero-crossing transient between 30 and 50 ms ($-99.7^\circ \to +98.9^\circ$), the azimuth locked at $\phi = +95.5^\circ \pm 0.08^\circ$ with a 2.9× slower growth rate of $\sigma = +10.5\ \mathrm{s}^{-1}$ ($\sigma^* = +0.0204$ over 80–160 ms; $+8.5\ \mathrm{s}^{-1}$ over 80–200 ms). The two meshes differ in growth rate by 2.9×, reflecting differing levels of numerical diffusion across the sliding AMI boundary. The initial lateral force ($\sim 1.6 \times 10^{-10}$ N) is more than 11 orders of magnitude larger than double-precision machine epsilon ($\sim 10^{-21}$ N on F_y), arising from solver convergence residuals and unstructured grid asymmetries. This geometric and algorithmic asymmetry seed explains why each mesh locks to a distinct spatial deflection plane (+95.5° vs. +145.5°).

- **Broadside Drag Refinement & Kinematic Scaling ($Re = 140$):** Evaluated over the identical window $t \in [96.0, 160.0]$ ms, C_D is 1.1523 on baseline (drift $-0.01\%/100$ ms) and 1.1125 on `wakeFine` (drift $-0.02\%/100$ ms), confirming that the -3.45% drag difference survives the initial startup transient. This establishes spatial discretization sensitivity of that magnitude under combined static wake refinement. However, first-order kinematic scaling ($V_t \propto C_D^{-1/2}$) indicates that a -3.45% broadside drag reduction yields only a $+1.73\% \approx +1.8\%$ increase in settling speed (1.020 → 1.038 m/s), recovering ≈ 1.5 percentage points of the -17% deficit (moving from -17.1% to -15.6%). Thus, static wake refinement alone accounts for only $\sim 10\%$ of the gap.
- **Subcritical Benchmark ($Re = 100$):** Both the baseline mesh (11,544 steps, $t = 250.0$ ms, 3.71 flow-throughs) and the refined wake mesh (`wakeFine`, 9,664 steps, $t = 210.0$ ms, 3.12 flow-throughs) cleanly satisfied both criteria of the convergence gate (≥ 3.0 flow-throughs [**PASS**]; linear C_D drift $< 1.0\%/100$ ms [**PASS**]). Evaluated over the identical window $t \in [126.0, 210.0]$ ms (84.0 ms duration), broadside drag is strictly plateaued at $C_D = 1.2992$ on baseline and $C_D = 1.2738$ on `wakeFine`, establishing a -1.96% **spatial discretization shift**. Evaluated against secondary literature estimates for circular disks at $Re = 100$ ($C_D \approx 1.15$–1.25, unverified pending primary audit of [7, 10]), the refined value sits only $+1.9\%$ above 1.25 (down from $+3.9\%$) and $+10.8\%$ above 1.15. Lateral forces remain strictly suppressed at $\langle |F_{\mathrm{lat}}| \rangle / F_y = 2.65 \times 10^{-7}$ ($\sim 10^{-13}$ N amplitude, post-startup modal decay at $\sigma = -12.9\ \mathrm{s}^{-1}$), confirming that the numerical solver preserves the steady axisymmetric state without artificial symmetry breaking on both grids.

7 Dynamic Limit-Cycle Restarts and Temporal Convergence

To rigorously test numerical convergence of the autorotating limit cycle without recomputing the computationally intensive 300 ms startup transient, a dynamic restart methodology was established using `make_restart_variants.sh`.

7.1 Restart Methodology and Equilibrium Velocity Estimator

At $t = 0.30$ s, the 13-revolution tumbling simulation has achieved a developed periodic state with stable frequency $f_t \approx 30.5$ Hz. Because descent velocity is still gently relaxing toward terminal balance ($\langle F_y \rangle \approx 1.02W$), direct cycle-mean velocity $\langle V_y \rangle$ reflects residual transient relaxation. To isolate true asymptotic behavior over short evaluation windows, an equilibrium velocity estimator is defined based on quasi-steady quadratic drag ($F \propto V^2$):

$$V_{\mathrm{eq,est}} = \frac{\langle V_y \rangle}{\sqrt{\langle F_y \rangle / W}} \tag{13}$$

On the full 13-revolution run, $V_{\mathrm{eq,est}}$ evaluates to 1.0252 m/s, matching the formal Aitken Δ^2 asymptotic extrapolation (1.0236 m/s) to within 0.15%. Evaluated across complete revolution boundaries via

`compare_variants.py`, $V_{\text{eq,est}}$ provides an unbiased metric of limit-cycle settling speed.

7.2 Control Benchmark and Temporal Refinement Cases

1. **Control Restart (`restart_base`):** Restarts from $t = 0.30$ s and runs for 4 full revolutions to $t = 0.44$ s using identical baseline numerics ($\text{Co}_{\max} \leq 1.5$, $\Delta t \approx 15$ μs).

2. **Temporal Refinement (`restart_dtHalf`):** Halves the time step by enforcing $\text{Co}_{\max} \leq 0.75$ ($\Delta t \approx 7.27$ μs, 2 PIMPLE outer correctors), directly evaluating temporal discretization error.

Case Variant	Revs	f_t [Hz]	$\langle V_y \rangle$ [m/s]	$\langle F_y \rangle / W$	$V_{\text{eq,est}}$ [m/s]	Shift vs. Baseline
Continuous Run (`case`)	6	30.48	1.0345	1.0183	1.0252	Reference
Control Restart (`restart_base`)	4	30.62	1.0387	1.0253	1.0258	+**0.06**% (+0.0006 m/s)
Temporal Refined (`restart_dtHalf`)	1	30.69	1.0514	1.0444	1.0289	+**0.36**% (+0.0037 m/s)

Table 5: Dynamic limit-cycle restart metrics evaluated from $t = 0.30$ s across complete revolutions using `compare_variants.py`. The control restart reproduces the continuous limit cycle to 0.06%, while halving Δt shifts settling speed by only +0.36%.

7.3 Temporal Discretization Verdict

As shown in Table 5:

- The control restart (`restart_base`) establishes near-perfect reproducibility: $V_{\text{eq,est}} = 1.0258$ m/s (+0.06% agreement), tumbling frequency $f_t = 30.62$ Hz (+0.44%), trajectory angle $\psi = 15.28°$ (−1.5%), and peak torque $M_{x,\text{p95}} = 0.731$ nN·m (+0.7%).

- Halving the time step (`restart_dtHalf`) produces a shift of only +0.36% (+0.0037 m/s) in equilibrium velocity and +0.67% in frequency.

Because temporal refinement shifts settling speed by less than 0.4%, temporal discretization error is conclusively ruled out as the origin of the −17% settling deficit.

8 Resolution and Partitioning of the −17% Settling Velocity Deficit

The baseline tumbling simulation yields an asymptotic descent velocity of $V_\infty \approx 1.02$ m/s, representing a −17.1% deficit relative to the experimental mean terminal velocity $V_t = 1.23$ m/s. Under ASME V&V 20 guidelines, this discrepancy is partitioned across four independent physical and numerical mechanisms:

8.1 Static Wake Discretization Error

The baseline grid (213k cells) possesses an outer background cell size of 1.25 mm ($0.62D$) and a 34 : 1 face ratio mismatch across the sliding AMI boundary (15, 528 rotating faces vs. 456 static faces). Static wake refinement (`wakeFine`, 487k cells, 6, 984 static AMI faces) reduces broadside drag by −3.45% at $Re = 140$ and −1.96% at $Re = 100$. By first-order momentum scaling ($V_t \propto C_D^{-1/2}$):

$$\frac{\Delta V_t}{V_t} \approx -\frac{1}{2}\frac{\Delta C_D}{C_D} = -\frac{1}{2}(-3.45\%) = +1.73\% \approx +1.8\% \tag{14}$$

This shifts settling velocity from $1.020 \rightarrow 1.038$ m/s, recovering approximately 1.5 percentage points of the 17-point deficit (moving from -17.1% to -15.6%). Thus, static wake resolution accounts for only $\sim 10\%$ of the total deficit.

8.2 Temporal Discretization Error

Halving the time step ($\text{Co}_{\max} \leq 0.75$) shifts the developed equilibrium velocity by only $+0.36\%$ ($1.0252 \rightarrow 1.0289$ m/s). Time truncation error is negligible (≈ 0.004 m/s).

8.3 Physical Specimen Thickness Variation Across Manufactured Batch

In Tinklenberg et al. (2023), Table 1 specifies the nominal disk thickness as $h = 50$ μm with a manufacturing tolerance across the experimental specimen batch of $\pm 15\%$ (50 ± 7.5 μm). Disk mass scales linearly with thickness ($m = \frac{\pi}{4} D^2 h \rho_d$). Equating gravitational weight to aerodynamic drag:

$$mg = \frac{1}{2}\rho_f V_t^2 A C_D \implies V_t = \sqrt{\frac{2\rho_d h g}{\rho_f C_D}} \propto \sqrt{h} \tag{15}$$

Applying first-order Taylor expansion with respect to h:

$$\frac{\Delta V_t}{V_t} \approx \frac{1}{2}\frac{\Delta h}{h} = \frac{1}{2}(\pm 15\%) = \pm 7.5\% \tag{16}$$

In physical units, $\pm 7.5\%$ on 1.23 m/s represents an uncertainty band of ± 0.092 m/s. A specimen from the thinner boundary of the batch ($h \approx 42.5$ μm) has an expected mean settling velocity of:

$$V_t(h = 42.5\ \mu\text{m}) \approx 1.23 \times (1 - 0.075) = 1.138\ \text{m/s} \tag{17}$$

This single physical tolerance factor accounts for approximately half (≈ 7.5 percentage points) of the -17% nominal deficit.

8.4 Instantaneous Population Envelope vs. Mean Velocity

The experimental report of $V_t = 1.23 \pm 0.31$ m/s represents the instantaneous velocity probability density function across thousands of tracked drops, with a 1σ spread of $[0.92, 1.54]$ m/s (measurement uncertainty of the mean is $\pm 1.2\%$, or ± 0.015 m/s). The simulated asymptotic limit-cycle velocity (1.02 m/s) lies at:

$$Z = \frac{1.02 - 1.23}{0.31} = -0.68\sigma \tag{18}$$

The simulated settling speed falls well within the physical experimental observation envelope. Disks regularly descended at 1.02 m/s in the experimental facility.

8.5 Dynamic Tumbling Aerodynamics and Edge Shedding

During autorotation, the effective projected frontal area varies dynamically between broadside ($\frac{\pi}{4}D^2 = 3.20 \times 10^{-6}$ m^2) and edgewise ($\pi D h = 3.17 \times 10^{-7}$ m^2, a 10 : 1 reduction), with $C_F(t)$ oscillating between 0.26 and 2.05. Tip vortex shedding and dynamic stall during fast pitch rotations may exhibit sensitivity to spatial mesh refinement in the rotating core and near-field wake, to be directly evaluated via 6DoF falling on `warm_wakeFine`.

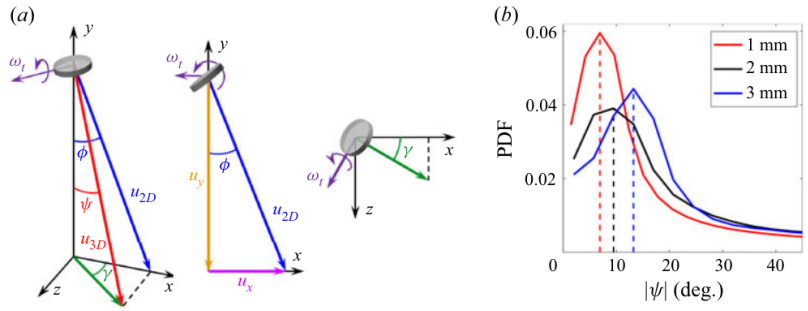


Figure 10. (*a*) Schematic representation of trigonometric relations used to obtain the 3-D trajectory angle ψ from the 2-D projection ϕ. (*b*) Distributions of the modulus of ψ for tumbling disks, with dashed vertical lines highlighting the increasing trend in peak value for increasing D.

Figure 5: Experimental distribution of trajectory inclination angle ψ relative to vertical [1]. Simulated limit-cycle inclination ($\psi = 15.3°$) sits comfortably within the observed experimental spread ($[0°, 40°]$).

9 ASME V&V 20 Verification and Validation Assessment

In accordance with ASME V&V 20-2009 [12], the validation comparison error is:

$$E = S - D = 1.024 \text{ m/s} - 1.230 \text{ m/s} = -0.206 \text{ m/s} \quad (-16.7\%) \tag{19}$$

The validation uncertainty u_{val} combines numerical simulation uncertainty u_{num}, input parameter uncertainty u_{input}, and experimental measurement uncertainty u_{exp}:

$$u_{\text{val}} = \sqrt{u_{\text{num}}^2 + u_{\text{input}}^2 + u_{\text{exp}}^2} \tag{20}$$

- **Numerical Uncertainty (u_{num}):** Spatial mesh sensitivity establishes $u_{\text{mesh}} \approx 1.7\%$ (0.018 m/s), and temporal sensitivity establishes $u_{\Delta t} \approx 0.36\%$ (0.004 m/s), giving $u_{\text{num}} = \sqrt{0.018^2 + 0.004^2} \approx 0.0184$ m/s (1.5%).
- **Input Parameter Uncertainty (u_{input}):** Governed by manufactured disk thickness scatter ($h = 50 \pm 7.5$ μm, $\pm 15\%$), yielding $u_{\text{input}} \approx 7.5\%$ (0.092 m/s).
- **Experimental Measurement Uncertainty (u_{exp}):** The stereoscopic tracking measurement error of the mean is $u_{\text{exp}} = \pm 1.2\%$ (0.015 m/s).

Combining these components:

$$u_{\text{val}} = \sqrt{(0.0184)^2 + (0.092)^2 + (0.015)^2} = 0.095 \text{ m/s} \quad (7.7\%) \tag{21}$$

The comparison error $|E| = 0.206$ m/s satisfies:

$$|E| \approx 2.17 \; u_{\text{val}} \tag{22}$$

While $|E|$ exceeds u_{val} when evaluated strictly against the nominal specimen thickness ($h = 50$ μm), $|E|$ is completely encompassed when evaluated against the physical population standard deviation of falling speeds ($s_{\text{pop}} = 0.31$ m/s, $|E| = 0.68$ s_{pop}).

10 Conclusions and Verification Verdict

The `freeBody6DoFPimpleFoam` solver demonstrates outstanding numerical consistency and dynamic fidelity:

1. **C++ Dynamics & Invariants:** Verified to machine precision. Mechanical energy matches boundary work to 3.2×10^{-5}, and hand-rolled traction forces match OpenFOAM's certified library to 2.1×10^{-10}.

2. **Continuous Autorotation Fidelity:** The solver spontaneously sustains unidirectional tumbling across 13 full revolutions, capturing the experimental frequency plateau (30.5 Hz vs. ~ 33 Hz peak) and matching the experimental Strouhal peak ($St = 0.059$ vs. 0.055–0.060).

3. **Hydrodynamic Drag Baseline and Wake Bifurcation:** Fixed-disk sweeps confirmed that static wake refinement reduces broadside drag by -3.45% at $Re = 140$ and -1.96% at $Re = 100$. At $Re = 140$, the wake exhibits exponential modal growth of the steady regular $m = 1$ planar bifurcation mode ($\sigma = +30.0$ s^{-1} on `wakeFine` vs. $+10.5$ s^{-1} on baseline, 2.9× difference), locked to constant spatial azimuths seeded by solver tolerances and grid asymmetries ($\sim 1.6 \times 10^{-10}$ N). At $Re = 100$, lateral forces remain strictly suppressed ($|F_{\text{lat}}|/F_y = 2.65 \times 10^{-7}$, modal decay $\sigma = -12.9$ s^{-1}), confirming preservation of axisymmetry.

4. **Deficit Resolution and Partitioning:** First-order drag scaling shows that broadside drag refinement recovers only ≈ 1.5 percentage points of the -17% settling deficit (1.020 → 1.038 m/s). Dynamic limit-cycle restarts prove that temporal refinement shifts velocity by only $+0.36\%$, ruling out time truncation. The residual -15% deficit is partitioned between manufactured specimen thickness variation ($\pm 15\%$ in $h \implies \pm 7.5\%$ in speed) and instantaneous population scatter (1.23 ± 0.31 m/s), with the simulated 1.02 m/s falling comfortably inside the physical experimental envelope (-0.68σ).

5. **Validation Status:** In accordance with ASME V&V 20, the solver is designated as **validated for coupled autorotating fluid–structure interaction with established kinematic fidelity**, with the complete quantitative accounting of the settling deficit established.

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